

A Sharper Focus: Clarifying the PFAS–Preeclampsia Association by Analyzing Disease Subtypes

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Per- and polyfluoroalkyl substances (PFAS) are sometimes called “forever chemicals” owing to their slow decay in the environment and long half-lives in people,¹ which for some PFAS exceeds a decade.² The chemicals have been associated with multiple health risks,¹ including adverse pregnancy outcomes, such as preterm birth, low birth weight, and preeclampsia.^{3,4,5,6,7} The clinical heterogeneity of preeclampsia has long been discussed⁸ but is rarely addressed in studies of environmental exposures.⁹ A study of PFAS recently published in *Environmental Health Perspectives*¹⁰ shows why future research should pay more attention to preeclampsia subtypes.

Preeclampsia, whose primary clinical feature is new-onset or worsening hypertension, affects 10–15% of worldwide pregnancies^{11,12} and is a major cause of maternal morbidity and mortality.¹³ “Preeclampsia is a syndrome with multiple independent or overlapping etiologies, and treating it as a single condition in analyses may miss associations with environmental factors,” says senior author David Cantonwine, an assistant professor of

epidemiology at Boston’s Brigham and Women’s Hospital. “That is why we tested if PFAS associations differed for subtypes defined by gestational age at diagnosis.”

The authors defined early-onset preeclampsia as receiving a diagnosis before 34 weeks of gestation and late-onset preeclampsia as diagnosed at 34 weeks or later.⁹ A clinical diagnosis of preeclampsia can be made after 20 weeks of gestation in women whose blood pressure was previously normal and who have protein in their urine.¹⁴ Early-onset preeclampsia is often associated with placental damage and fetal growth restriction, whereas these features are less common in the late-onset form.^{15,16,17}

The researchers analyzed 75 cases and 75 controls (mean age 32.2 years) randomly chosen from 1,648 participants in the LIFECODES cohort¹⁸ between 2006 and 2008. Plasma levels of two biomarkers—placental growth factor and soluble fms-like tyrosine kinase-1—had previously been measured four times during pregnancy (median 10, 18, 26, and 35 weeks).¹⁸ The two biomarkers are used to estimate the balance of pro- and anti-



The authors divided preeclampsia into early- and late-onset subtypes. They found that higher plasma concentrations of two PFAS in early pregnancy were associated with greater odds of late-onset preeclampsia but not the early-onset subtype. These results indicate that subtypes of preeclampsia should be considered in future studies of the disease. Image: © Rawpixel.com/stock.adobe.com.

angiogenic activity within the placenta,¹⁹ which may be influenced by PFAS.^{20,21}

The researchers measured nine PFAS in plasma samples collected early in pregnancy (median 10 weeks). Early measurements are less likely to be confounded by preeclampsia-related changes in kidney function later in pregnancy^{10,22} and to have a greater potential to inform risk assessment and clinical care.²³

The authors reported that higher concentrations of two PFAS in early pregnancy were associated with greater odds of late-onset preeclampsia but did not change the odds of the early-onset subtype. On the other hand, early PFAS concentrations were not robustly associated with changes in the two angiogenic biomarkers during the course of pregnancy. Because these biomarkers may better predict early-onset preeclampsia,^{24,25} this finding supports the hypothesis that PFAS may influence the late-onset subtype via other mechanisms.

“We think the different PFAS associations indicate that late-onset preeclampsia may be due to mechanisms other than disruptions in angiogenesis, such as dyslipidemia and altered thyroid hormone functioning,” says Cantonwine.

Most early-onset preeclampsia cases, he adds, show consistent signs of early ischemic damage to the placenta. Less pronounced damage may develop later in some, but not all, women diagnosed after 34 weeks of gestation.

For Tracey Woodruff, a professor of reproductive sciences at the University of California, San Francisco, these mechanistic differences between preeclampsia subtypes are a plausible explanation for the observed data and highlight the importance of accounting for gestational age at diagnosis. “Incorporating clinical biomarkers into an analysis of environmental chemicals is another strength of the study,” says Woodruff, who was not involved in the project. “Using these biomarkers to tease out potential etiologic differences between subtypes is valuable for future studies.”

Tracy Manuck, an associate professor of obstetrics and gynecology at the University of North Carolina at Chapel Hill, who also was not involved in the project, agrees that the joint analysis of well-known biomarkers and PFAS levels is a unique contribution to the field. “Despite the small sample size, I think the findings are very interesting and show that precise phenotyping may help identify associations between environmental chemicals and adverse pregnancy outcomes,” she says.

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